CHEMISTRY

1)

Arrange the following elements in increasing order of first ionization enthalpy :
Li, Be, B, C, N
Choose the correct answer from the options given below :

(1) Li < Be < B < C < N

(2) Li < B < Be < C < N

(3) Li < Be < C < B < N

(4) Li < Be < N < B < C

2) Identify the **incorrect** statement :

(1) Na_2O is a basic oxide and Cl_2O_7 is an acidic oxide.

(2) The following four species are called isoelectronic species : O^{2-}, F^-, Na^+ and Mg^{2+}.

(3) Among the four species Mg, Al, Mg^{2+} and Al^{3+}, the smallest one is Al.

(4) The oxidation and coordination number (or covalency) of Al in $[AlCl(H_2O)_5]^{2+}$ are +3 and 6, respectively

3)

Arrange the following elements in increasing order of electronegativity.
N, O, F, C, Si
Choose the correct answer from the options given below :

(1) Si < C < N < O < F

(2) Si < C < O < N < F

(3) O < F < N < C < Si

(4) F < O < N < C < Si

4) **Assertion :-** Al_2O_3 and As_2O_3 behave as acidic with bases and as basic with acids
Reason :- Al_2O_3 and As_2O_3 are amphoteric oxides

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**

(2) **Assertion** is true but **Reason** is false

(3) **Assertion** is false but **Reason** is true

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**

5) **Statement-I :** According to modern periodic law, " the physical and chemical properties of elements are periodic functions of their atomic mass."
Statement-II : In long form of periodic table the horizontal rows (which Mendeleev called

series) are called periods and the vertical coloums groups.

(1) Both statement I and statement II are true.
(2) Both statement I and statement II are false.
(3) Statement I is true but statement II is false.
(4) Statement I is false but statement II is true.

6) **Statement-I :** Neptunium and Plutonium like Actinium and Protactinium are found in pitche blend i.e., an ore of uranium.
Statement-II : A plot of $\sqrt{\nu}$ against atomic mass gave a straight line (where ν is frequency of x-rays emitted).

(1) Both statement I and statement II are true.
(2) Both statement I and statement II are false.
(3) Statement I is true but statement II is false.
(4) Statement I is false but statement II is true.

7) Incorrect order of radius is :-

(1) Li < Be < B < C
(2) Na > Al > P > Cl
(3) $N^{3-} > O^{2-} > F^{-} > Na^{+}$
(4) $S^{2-} > Cl^{-} > K^{+} > Ca^{2+}$

8) Which inert gas doesn't show ns^2, np^6 configuration

(1) Og
(2) H
(3) Ar
(4) He

9) Actinoid elements are

(1) Z = 57 to 71
(2) Z = 90 to 103
(3) Z = 89 to 103
(4) Z = 58 to 71

10) Wrong match is :-

(1) Z = 101 = Mendelivium
(2) Z = 106 = Seaborgium
(3) Z = 118 = Tennessine
(4) Z = 115 = Moscovium

11) p-block contains :-

(1) metals
(2) non-metals
(3) metalloids
(4) all

12) Wrong order of electronic configuration is

(1) 6s, 4f, 5d, 6p
(2) 1s, 2s, 2p, 3s, 3p, 3d
(3) 6p, 7s, 5f, 6d, 7p
(4) 4s, 3d, 4p, 5s, 4d

13) Which of the following is an incorrect match

(1) Representative element : Li & C elements
(2) Transition element : all d-block elements
(3) Inert gas : 18 group elements
(4) Halogen : 17 group elements

14) Which of the following can't be an atomic number of an inert gas

(1) 86
(2) 52
(3) 36
(4) 10

15) The radius of N, N^{-3}, O and O^{-2} are in the order of :-

(1) $N^{-3} > O^{-2} > O > N$
(2) $O^{-2} > N^{-3} > O > N$
(3) $N^{-3} > N > O^{-2} > O$
(4) $N^{-3} > O^{-2} > N > O$

16) Which of the following sets represents the collection of isoelectronic species ?

(1) Na^{+}, Mg^{+2}, Al^{+3}, Cl^{-}
(2) Na^{+}, Ca^{+2}, Sc^{+3}, F^{-}
(3) K^{+}, Cl^{-}, Mg^{+2}, Sc^{+3}
(4) K^{+}, Ca^{+2}, Sc^{+3}, Cl^{-}

17) Which of the following trend is wrong for radius :-

(1) Ni < Cu < Zn
(2) Fe ≈ Co ≈ Ni
(3) Ti < Zr ≃ Hf

(4) Li < Be < B

18) The second ionisation energy is maximum for :

(1) Ba
(2) Be
(3) Mg
(4) B

19) The values of successive I.P.'s of an element are 7.1, 14.3, 34.5, 48.6, 162.2 eV respectively. The element will be-

(1) Na
(2) Se
(3) F
(4) Si

20) Which process requires absorption of energy ?

(1) $Cl \rightarrow Cl^-$
(2) $F \rightarrow F^-$
(3) $O \rightarrow O^-$
(4) $N \rightarrow N^{3-}$

21) Correct order of acidic strength is-

(1) $Cl_2O_7 < SO_3 < P_4O_{10}$
(2) $Na_2O > MgO > Al_2O_3$
(3) $K_2O > CaO > MgO$
(4) $P_2O_5 < SO_3 < Cl_2O_7$

22) In which of the following arrangement the order is not according to the property indicate against it -

(1) Sc < Y < La (Atomic size)
(2) C < B < N < O (2nd Ionisation enthalpy)
(3) I < Br < F < Cl (Increasing electron gain enthalpy with negative sign)
(4) B > Al > Ga > In > Tl (Electronegativity)

23) Which statement/s is/are true.
(i) 1st I.P. of an element is always an endothermic process.
(ii) 1st E.A. of an element is always an exothermic process.
(iii) Generally EN of elements increases across the period.
(iv) Basic character of oxides increases down the group for I A and II A elements.

(1) (i) only

(2) (ii) and (iii)

(3) (i) and (iii)

(4) (i), (iii) and (iv)

24) Correct arrangement of following elements in increasing order of metallic character is -
Si, Be, Mg, Na, P

(1) P < Si < Be < Mg < Na

(2) P < Si < Mg < Na < Be

(3) Be < Si < P < Mg < Na

(4) Be < Si < P < Na < Mg

25) An element "X" with electronic configuration "$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$" is present in -

(1) 5th period & 5th group

(2) 5th period & 6th group

(3) 4th period & 5th group

(4) 4th period & 6th group

26) **Assertion :-** Atoms and ions which contains the same number of electrons are called isoelectronic species.
Reason :- Isoelectronic species have same number of protons.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion

(3) Assertion is true but Reason is false

(4) Both Assertion and Reason are false

27) Identify the incorrect match :-

	IUPAC name		**Elemental name**
(a)	Unnilunium	(i)	Copernicium
(b)	Unununium	(ii)	Lawrencium
(c)	Unnilpentium	(iii)	Rontgenium
(d)	Ununbium	(iv)	Mendelevium

(1) b-iii

(2) c-ii

(3) a-iv

(4) d-i

28) Match the column-I with column-II and select the correct answer using given codes :-

	Column-I (Number of periods)		Column-II (Number of elements)
(A)	4th period	(p)	14
(B)	2nd period	(q)	8
(C)	Lanthanides	(r)	32
(D)	3rd group	(s)	18

(1) (A-p), (B-r), (C-s), (D-q)

(2) (A-q), (B-r), (C-s), (D-p)

(3) (A-s), (B-q), (C-p), (D-r)

(4) (A-s), (B-p), (C-q), (D-r)

29) **Assertion :-** Group 17 elements (Halogens) have very high negative electron gain enthalpies.
Reason :- They can attain stable noble gas electronic configuration by gaining an electron.

(1) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(2) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

30) Arrange the following species in the increasing order of ionic radius.
N^{-3}, O^{-2}, F^{-}, Na^{+}, Mg^{+2} and Al^{+3}

(1) $N^{-3} < O^{-2} < F^{-} < Na^{+} < Mg^{+2} < Al^{+3}$

(2) $N^{-3} = O^{-2} = F^{-} = Na^{+} = Mg^{+2} = Al^{+3}$

(3) $Al^{+3} < Mg^{+2} < Na^{+} < F^{-} < O^{-2} < N^{-3}$

(4) $O^{-2} < F^{-} < Na^{+} < Mg^{+2} < Al^{+3} < N^{-3}$

31) The elements Z = 117 have been discovered. Then electronic configuration and group number are :-

(1) [Rn] $5f^{14}$, $6d^{10}$, $7s^{2}$, $7p^{5}$ Group-17

(2) [Xe] $5f^{14}$, $6d^{10}$, $7s^{2}$, $7p^{3}$ Group-17

(3) [Rn] $5f^{14}$, $6d^{10}$, $7s^{0}$, $7p^{5}$ Group-15

(4) [Xe] $4f^{14}$, $5d^{10}$, $6s^{2}$, $6p^{5}$ Group-13

32) General electronic configuration of d-block elements are represent by :-

(1) $(n-1)\ d^{1-10}\ ns^{2}$

(2) $(n-2)\ f^{1-14}\ (n-1)d^{0-1}ns^{2}$

(3) $(n-1)\ s^{2}, p^{6}, d^{1-10}\ ns^{0-2}$

(4) $ns^{2}\ np^{1-6}$

33) IUPAC symbol and IUPAC official name of atomic number 108 :-

(1) Uns, Bohrium

(2) Uno, Hassium

(3) Uuo, Hassium

(4) Uon, meitnerium

34) Given values of ionisation enthalpy of element X, then the electronic configuration of element X is (IE in eV) :-

IE_1	IE_2	IE_3	IE_4	IE_5
32	65	700	900	1000

(1) $1s^2\ 2s^2$

(2) $1s^2\ 2s^2\ 2p^6\ 3s^2$

(3) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

(4) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

35) How many statements are incorrect :
(1) Cs has least ionisation enthalpy
(2) O is more electronegative than N
(3) Electronegativity is property of isolated atom
(4) Ionic radii order $Ca^{2+} > K^+$
(5) Genarally ionisation enthalpy increases with decrease in size.

(1) 1

(2) 2

(3) 3

(4) 4

36) Reasons for the different chemical behaviour of the first element of group 1 and 2 compared to that of the other members of their respective group are :-
(i) Its small size
(ii) Its large (charge/radius) ratio
(iii) High electronegativity of the element

(1) (i) only

(2) (i) and (ii) only

(3) (i) and (iii) only

(4) (i), (ii) and (iii)

37) **Assertion :-** Li and Mg show diagonal relationship.
Reason :- Li and Mg are alkali metal and alkaline earth metal respectively.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion

(3) Assertion is true but Reason is false

(4) Both Assertion and Reason are false

38) The element with higest negative electron gain enthalpy will belong to :-

(1) Period 2, group 18
(2) Period 2, group 17
(3) Period 3, group 17
(4) Period 2, group 1

39)

The first ionisation enthalpies (kJ mol^{-1}) of Mg and Al respectively are.

(1) 801, 899
(2) 899, 801
(3) 899, 899
(4) 801, 801

40) **Assertion :** In a boron family, down the group electronegativity regularly decreases.
Reason : Atomic size of Ga is smaller to size of Al.

(1) Assertion and Reason both are correct but Reason is not correct explanation of Assertion.
(2) Assertion is incorrect but Reason is correct.
(3) Assertion is correct but Reason is incorrect.
(4) Assertion and Reason both are correct and Reason is correct explanation of Assertion.

41) Which of the following represent correct order of electron gain enthalpy.

(1) P > S > N > O
(2) S > O > P > N
(3) S > Se > O
(4) Both (2) and (3)

42) Correct electronic configuration of atomic number 115 is :-

(1) [Xe] $6s^2$, $4f^{14}$, $5d^{10}$, $6p^3$
(2) [Rn] $7s^2$, $5f^{14}$, $6d^{10}$, $7p^3$
(3) [Rn] $7s^2$, $5f^{14}$, $6d^{10}$, $7p^1$
(4) [Xe] $6s^2$, $4f^{14}$, $5d^{10}$, $6p^2$

43) Which of the following is correct for inert gases in increasing order of positive electron gain enthalpy ($\Delta_{eg}H$)

(1) He < Ne < Ar < Kr < Xe < Rn
(2) Rn < Xe < Kr < Ar < Ne < He
(3) He < Rn < Xe < Ar = Kr < Ne
(4) He < Rn < Xe < Kr < Ar < Ne

44) Which of the following statement is incorrect ?

(1) Boron-11 (B^{11}) isotope has high ability to absorb neutron.

(2) ^{14}C is a radioactive isotope with half life 5770 years and used for radiocarbon dating.

(3) The electronegativity values for elements from Si to Pb are almost the same.

(4) Lead donot show catenation.

45) The tendency to show –3 oxidation state decreases down the group in group 15 due to

(1) Increase in size

(2) decrease in metallic character

(3) Increase in metallic character

(4) Both (1) and (3)

46) **Assertion (A) :** Actinoid contraction is greater than lanthanoid contraction from element to element.
Reason (R) : The 5f electron's shielding is more poor than that of 4f electrons
Correct option is -

(1) A is correct but R is incorrect

(2) A is incorrect but R is correct

(3) Both A and R are correct and R is correct explanation of A

(4) Both A and R are correct but R is not correct explanation of A

47) An element has electronic configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$. Predict their period, group and block.

(1) Period = 3rd, Block = p, Group = 15

(2) Period = 5th, Block = s, Group = 1

(3) Period = 3rd, Block = p, Group = 10

(4) Period = 4th, Block = d, Group = 12

48) Select the correct statements is/are -
(A) Ionisation enthalpy is always negative (for a neutral atom)
(B) The second ionisation enthalpy will be higher than the first ionisation enthalpy.
(C) Ionisation enthalpy of alkali metals are highest in each period.
(D) Ionisation enthalpy of noble gases are highest in each period.

(1) (A), (C) are correct

(2) (B), (D) are correct

(3) (A), (B), (C), (D) are correct

(4) (C), (D) are correct

49)

What would be the correct e– configuration of element $_{64}Gd$?

(1) $[Xe]4f^8 5d^0 6s^2$

(2) $[Xe]4f^{7}5d^{3}6s^{0}$

(3) $[Xe]4f^{7}5d^{1}6s^{2}$

(4) $[Xe]4f^{6}5d^{2}6s^{2}$

50) Read the following statement :
(A) Americans named of atomic number (Z = 103) is Rutherfordium whereas soviets named it kurchatovium.
(B) 'Tl' is p-block element.
(C) Radius of Li is greater than I.
(D) A cation is smaller than its parent atom because it has fewer electrons while its nuclear charge remains the same.
How many total no. of above statements are correct :

(1) 2

(2) 3

(3) 1

(4) 4

51) Acidic oxide amongst the following.

(1) B_2O_3

(2) Al_2O_3

(3) Ga_2O_3

(4) In_2O_3

52) Which of the following order is incorrect ?

(1) $Ce^{3+} > Pr^{3+} > Nd^{+3} > Pm^{+3}$ (Ionic radii)

(2) $Na^{+} > Li^{+} > Mg^{2+} > Be^{2+}$ (Ionic radii)

(3) F > Cl > Br > I (Ionization energy)

(4) P > S < Cl (Electron affinity)

53) If IE_1 of O^{-} ion is equal to '+ X' ev/atom then ΔH_{eg} of oxygen atom is equal to :-

(1) – X ev

(2) + 2X ev

(3) + 'X' ev

(4) – '2X' ev

54) Number of electrons in 3rd shell of Pd are :-

(1) 20

(2) 25

(3) 18

(4) 30

55) The ionization energy will be minimum and maximum respectively for the process :-

(1) $Ba_{(g)} \rightarrow Ba^{+2}_{(g)}$ (2) $Be_{(g)} \rightarrow Be^{+2}_{(g)}$

(3) $Cs_{(g)} \rightarrow Cs^{+}_{(g)}$ (4) $Li_{(g)} \rightarrow Li^{+}_{(g)}$

(1) 1 and 2
(2) 3 and 2
(3) 3 and 4
(4) 1 and 4

56) Which of the following property increases regularly from carbon to fluorine ?

(1) I.P.
(2) E.A.
(3) Shielding constant
(4) All of these

57) Which of the following is correct size order :-

(1) C < O < S < Se
(2) O < C < S < Se
(3) C < S < O < Se
(4) O < C < Se < S

58) Which element of IIA is smaller than potassium

(1) Sr
(2) Ba
(3) Mg
(4) All of these

59) Correct order of second ionisation potential of C, N, O and F is :

(1) C > N > O > F
(2) O > F > N > C
(3) O > N > F > C
(4) F > O > N > C

60) Select the most basic hydride ?

(1) NaH
(2) MgH_2
(3) AlH_3
(4) RbH

61) The screening effect of 'd' electrons is :

(1) Much less than s-electrons
(2) Much more than s-electrons
(3) Equal to s-electrons
(4) Equal to p-electrons

62) The decreasing order of ionic character of N–H, F–H, C–H and O–H is :-

(1) N–H > F–H > C–H > O–H
(2) F–H > N–H > C–H > O–H
(3) O–H > C–H > F–H > N–H
(4) F–H > O–H > N–H > C–H

63) In which molecule does the chlorine atom has positive partial charge :-

(1) HCl
(2) BrCl
(3) OCl_2
(4) SCl_2

64) The element X, Y, Z and J have the indicated electron configurations starting with the innermost shell. The most metallic element is :-

(1) X = 2, 8, 3
(2) Y = 2, 8, 8
(3) Z = 2, 8, 8, 1
(4) J = 2, 8, 8, 7

65) Incorrect order of IP is :

(1) $Cu^+ > Zn^+$
(2) $K^+ < Cu^+$
(3) K < Cu
(4) Cu < Zn

66) Which of the following have different value of electronegativity :-

(1) H
(2) S
(3) C
(4) I

67) Number of elements in 4th period which has atleast one e^- in d subshell ?

(1) 10
(2) 12
(3) 16

(4) 18

68) The option with only amphoteric oxide is/are :-

(1) Cr_2O_3, B_2O_3, SnO, SnO_2

(2) Cr_2O_3, CrO, SnO, PbO_2

(3) NO, B_2O_3, PbO, SnO_2

(4) ZnO, Al_2O_3, PbO, PbO_2

69) **Assertion :** Cd is a transition element.
Reason : Cd and Cd^{+2} both have completely filled d-orbital in their ground state.

(1) **Assertion** is true but **Reason** is false.

(2) **Assertion** is false but **Reason** is true.

(3) Both **Assertion** and **Reason** are true and **Reason** is correct explanation for **Assertion**.

(4) Both **Assertion** and **Reason** are true but **Reason** is not correct explanation for **Assertion**.

70) Which of the following configuration is associated with biggest jump between 2nd and 3rd I.E. ?

(1) $1s^2$, $2s^22p^2$

(2) $1s^2$, $2s^22p^6$, $3s^1$

(3) $1s^2$, $2s^22p^6$, $3s^2$

(4) $1s^2$, $2s^22p^1$

71) Group-13 elements react with O_2 in amorphous form to form oxides of type M_2O_3 (M = element). Which among the following is the most basic oxide?

(1) Al_2O_3

(2) Ga_2O_3

(3) Tl_2O_3

(4) B_2O_3

72) Calculate total number of electrons of 5th, 4th and 3rd shell in 'Tc'

(1) 34

(2) 33

(3) 35

(4) 36

73) Which of the following can react with NaOH ?

(1) Rb_2O

(2) CaO

(3) SO_3

(4) KOH

74)

Which of the following is typical element of group 14 ?

(1) C
(2) Al
(3) Si
(4) P

75) In which of the following compound 'Xe' show's maximum Electronegativity ?

(1) XeF_2
(2) $XeOF_2$
(3) XeO_3
(4) XeO_4

ANSWER KEYS

CHEMISTRY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**2**	**3**	**1**	**4**	**4**	**3**	**1**	**4**	**2**	**3**	**4**	**2**	**2**	**2**	**4**	**4**	**4**	**4**	**4**	**4**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**4**	**4**	**4**	**1**	**4**	**3**	**2**	**3**	**2**	**3**	**1**	**3**	**2**	**2**	**2**	**4**	**2**	**3**	**2**	**2**
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	**4**	**2**	**3**	**1**	**4**	**3**	**1**	**2**	**3**	**2**	**1**	**4**	**1**	**3**	**2**	**3**	**2**	**4**	**2**	**4**
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75					
A.	**1**	**4**	**3**	**3**	**2**	**1**	**3**	**4**	**2**	**3**	**3**	**2**	**3**	**3**	**4**					

SOLUTIONS

CHEMISTRY

1) Li < Be > B < C < N
$2s^1$ $2s^2$ $2s^22p^1$ $2s^22p^2$ $2p^22p^3$
Li < B < Be < C < N

2)

Correct order of radius – $Al^{+3} < Mg^{+2} < Al < Mg$
Al in $[AlCl(H_2O)_5]^{2+}$ are +3 and 6, respectively

3) EN order (Pauling scale value)
Si < C < N < O < F
1.8 < 2.5 < 3.0 < 3.5 < 4

4) Amphoteric oxides can behave as acid as need as base.

5) atomic number not atomic mass.

6) $\sqrt{\nu}$ v/s Z graph ⇒ straight line.
(Z = at. No.)

7) (i) Be > B > C > N > O > F (r)

8) He = $1s^2$

9) An = 90 to 103

10) Z = 118 = oganesson Og

11) p-block contains, metal non metals, metalloid all kind of elements

12) 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p

13) all d-block other than Zn, Cd, Hg and Cn.

14) 2^{He},10^{Ne},18^{Ar},36^{Kr}, 54^{Xe}, 86^{Rn}, 118^{Og}

15)

$N^{-3} > O^{-2}$ isoelectronic –ive ↑ size ↑
N > O Z_{eff} ↑ size ↓

16)

$K^{+} = 18$
$Ca^{+2} = 18$
$Sc^{+3} = 18$
$Cl^{-} = 18$

17)

L → R in period
Z_{eff} ↑ size ↓

18)

$$\text{I.P.} \propto \text{zeff} \propto \frac{1}{\text{At. Size}}$$

19)

There is a high jump of I.P. values after 4th value of I.P. thus element contain 4 valence electrons in its valence shell.

20)

Formation of polynegative anions is always an endothermic process due to repulsion between anion and electron.

21)

Acidic strength of oxides ∝ O.N. of element
Acidic strength ∝ EN

22)

The order of E.N. for boron family is.
B > Tl > In > Ga > Al

23)

1st EA may be exothermic or endothermic in nature.

24) Metallic character increases top to bottom and decreases left to right.

25)

$_{24}Cr \Rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$
Group Number = 6 (1 + 5)
Period number = 4

26)

NCERT Pg. # 84

27)

NCERT Pg. # 77

28)

NCERT-XIth Pg. 75 (Part-I)

29)

NCERT-XI Pg. # 86 (Part-I)

30)

NCERT-XI Pg. # 93 (Part-I)
$\left(\text{Ionic radius} \propto \frac{-ve}{+ve}\right)$

31)

NCERT-XI Pg. # 81 (Part-I)
Halogen family (Group-17)

32)

NCERT-XI Part-I Pg. # 81

33)

NCERT-XI Part-I Pg. # 77
Uno, Hs

34)

After removal of $2e^-$ element x gain noble gas configuration.

35) EN is property of bonded state
Ionic radius $K^+ > Ca^{+2}$

36)

NCERT XI, Pg. # 94

37)

NCERT XI, Pg. # 90

38)

NCERT Pg. # 90

39)

NCERT Pg. # 88

40)

NCERT, Pg. # 88

41)

NCERT-XI Pg. # 87 (Part-I)

42)

Z = 115 [Group 15]
GEC = $ns^2\ np^3$

43)

$\Delta_{eg}H$ (+ve) value ⇒
He = +48, Ne = +116, Ar = +96, Kr = +96,
⇒ Xe = + 77, Rn = +68

44)

NCERT XI Pg. # 112

45)

NCERT-XII Pg.#172

46)

NCERT Pg. No. # 239

47)

NCERT Pg # 83

48) $IP_1 < IP_2$

49) $_{64}Gd = {}_{51}[Xe]4f^7 5d^1 6s^2$

50)

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52) P < S < Cl (Order of E.A.)

57)

Se = 4th period

S = 3rd period
C, O = 2nd period
C > O = (Z_{eff} ↑)

66) C = S = I = 2.5 (E.N.)

67)

In 4th period only K and Ca do not have e^- in d subshell.
So total element which = 18 – 2 = 16 has d e^-

72)

$5s^2$
$4s^2 4p^6 4d^6$
$3s^2 3p^6\ 3d^5$ = 33 e^-

73) NaOH (Basic) + SO_3 (Acidic)

74) 3 Period elements → Typical elements